

HIGH

Deepfake CEO Fraud

Alert ID: ALERT-DE | Generated: 2024-06-13 14:45:00 UTC

Routed to: Compliance Team · Fraud Risk Team · Policy Controls Team

EXECUTIVE SUMMARY

A HIGH-priority AI Deepfake CFO Fraud threat has been identified, directly relevant to UOB's Trade Finance and SME operations in Singapore and Hong Kong. The February 2024 Hong Kong incident — USD 25M lost in a single deepfake video call — demonstrates the scale of potential impact for UOB's 11,000 trade finance customers. Current controls detect only 6% of these attacks, with an estimated unmitigated exposure of SGD 47M (P50) to SGD 125M (P90). Immediate action: implement dual-authorisation for all wire transfers exceeding SGD 50,000 and issue advisory to Trade Finance relationship managers.

THREAT INTELLIGENCE BRIEF

AI-powered deepfake technology has advanced to the point where criminals can generate real-time, convincing video and audio impersonations of corporate executives during live video conference calls. The attack sequence: (1) Criminal researches target company executives via LinkedIn/public records → (2) Schedules video call with finance team posing as CFO/CEO → (3) Uses real-time AI deepfake to appear and sound convincingly as the executive → (4) Instructs finance employee to process urgent wire transfer → (5) Funds transferred to criminal-controlled account before detection. The February 2024 Hong Kong case (USD 25M / SGD 33.7M) involved a multi-participant call with several deepfaked executives, bypassing the social engineering skepticism of a single-participant call. ENISA identifies Trade Finance, Correspondent Banking, and large SMEs as primary targets due to the scale of authorized transactions.

UOB RELEVANCE ASSESSMENT

Gate	Result	Details
Gate 1 — Novelty	✓ PASSED	Very high novelty (0.88): real-time AI deepfake in live video calls is a fundamentally new attack capability with no ex
Gate 2 — Customer Exposure	✓ PASSED	✓ TRADE_FINANCE customers in SG/HK; ~11,000 at risk (lower count but extremely high per-victim loss)
Gate 3 — Control Gap	✓ PASSED	✓ 85% undetected; no TM rule specifically targets deepfake-authorized transfers

Composite Risk Score: 74/100 — HIGH Priority

UOB's Trade Finance division processes large cross-border wire transfers — exactly the transaction profile targeted. Current controls (TM-008: invoice discrepancy detection) provide only 6% coverage against this attack, as the transfers appear legitimate (employee-authorized). 11,000 Trade Finance and SME corporate customers are estimated at risk. With avg loss potential of SGD 250,000 per incident, even 10 successful attacks would cost SGD 2.5M. Novelty score: 0.88 — this is a genuinely new threat capability requiring new countermeasures. Composite risk score: 73.8/100 (HIGH).

FINANCIAL IMPACT SIMULATION

Scenario	Exposure (SGD)
Baseline (no intervention)	SGD 55.0M
With current TM controls	SGD 51.7M
With proposed interventions	SGD 17.6M
P10 (optimistic)	SGD 18.0M
P50 (median)	SGD 47.0M
P90 (worst case)	SGD 125.0M

Baseline exposure: SGD 55M. With current controls: SGD 51.7M (invoice anomaly detection provides only 6% coverage). With proposed interventions (dual-auth + callback): SGD 17.6M. P10: SGD 18M, P50: SGD 47M, P90: SGD 125M.

RECOMMENDED ACTIONS

■ Immediate (<24 hours)

- Issue advisory to all Trade Finance relationship managers on deepfake video call threat with HK incident case study
- Flag all wire transfers > SGD 100,000 for manual review until dual-authorisation policy is deployed
- Brief corporate banking operations team on verification protocol for large-value transfer requests

■ Short-term (1–7 days)

- Implement dual-authorisation for wire transfers > SGD 50,000 (est. 40% exposure reduction)
- Establish mandatory callback verification protocol for transfers > SGD 100,000
- Deploy TM detection rule: large wire + video call + new payee within 4 hours
- Update corporate customer security advisory with deepfake verification guidance
- Brief trade finance operations team on how to independently verify executive identity

■ Strategic (1–4 weeks)

- Commission AI deepfake detection capability assessment (biometric liveness detection on video calls)
- Develop formal Deepfake Fraud Incident Response Protocol with CISO and Legal sign-off

PROPOSED INTERVENTION RULES

1. TM DETECTION Risk: MEDIUM · Impact: ~30% · Auto-deployable: No

Flag large wire transfers authorized immediately after video conferencing application activity on the approver's registered device.

```
IF wire_transfer > SGD 50000 AND approver_device_has_recent_video_call_activity AND
payee_is_new AND time_since_video_call < 4h THEN flag DEEPFAKE_CF0_RISK; hold 30 minutes;
escalate to Operations Manager
```

■ Requires Head of Fraud Risk sign-off

2. CUSTOMER FRICTION Risk: MEDIUM · Impact: ~40% · Auto-deployable: No

Implement mandatory dual-authorisation for all outward wire transfers exceeding SGD 50,000, requiring separate approval from a second authorised signatory.

```
IF transfer_type = OUTWARD_WIRE AND amount > SGD 50000 THEN require secondary_authorisation
from Operations_Manager_or_above; timeout = 2h; on timeout, reject and notify Compliance
```

■ Requires Head of Fraud Risk sign-off

3. POLICY_CHANGE Risk: LOW · Impact: ~35% · Auto-deployable: No

Mandate out-of-band callback verification via pre-registered executive hotline for all wire transfers > SGD 100,000 authorized via digital channels.

```
IF transfer_amount > SGD 100000 AND channel IN [INTERNET_BANKING, MOBILE] THEN operations_team
MUST call pre-registered_executive_number to verbally confirm before processing; record call;
retain for 7 years
```

■ Requires Head of Fraud Risk sign-off

4. ADVISORY Risk: LOW · Impact: ~20% · Auto-deployable: Yes

Issue advisory to Trade Finance relationship managers and corporate customers explaining deepfake video call threat and how to verify executive authenticity.

```
IF customer_segment IN [TRADE_FINANCE, SME] THEN send deepfake_advisory to relationship_manager
+ corporate_finance_contact with verification_protocol_guide WITHIN 24h
```

EVIDENCE SOURCES

1. ENISA Threat Landscape Report — AI-Powered Financial Fraud 2024
2. Hong Kong Police Force — February 2024 Deepfake CFO Fraud Case (USD 25M)
3. FBI Alert AC-000159-MW — Business Email Compromise / Deepfake Evolution

ANALYST NOTES & CAVEATS

This alert is AI-generated. It must be reviewed by a qualified fraud risk analyst before operational action is taken. Financial exposure figures assume USD 25M per major incident as reference point from Hong Kong case; actual UOB trade finance transaction values may differ. The P90 scenario (SGD 125M) assumes multiple successful attacks at trade finance transaction scale. Recommended validation: confirm dual-authorisation status of current Trade Finance wire approval workflows with Operations.